

Matrices & vectors

A column vector is an $n \times 1$ matrix $\rightarrow \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$ column vector

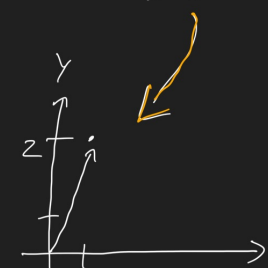
A row matrix is a $1 \times n$ matrix $\rightarrow (0, 0)$
 $\langle 1, 2, 3 \rangle$ or $(1 \ 2 \ 3)$

An n -dimensional vector is a vector w/ n entries

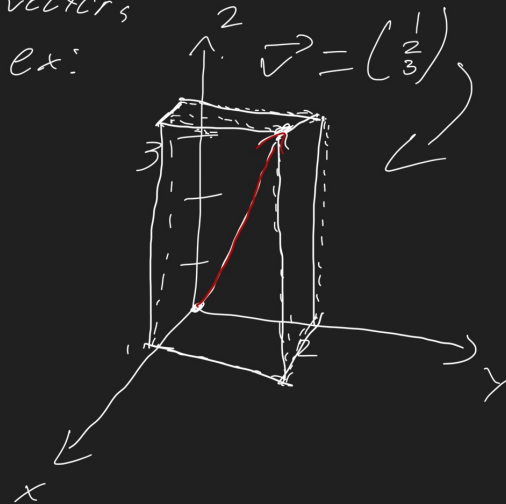
ex: $\mathbb{R} = \left\{ \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} : a_1, a_2, \dots, a_n \in \mathbb{R} \right\}$
 all real #'s

Note $\mathbb{C}^n$ set of all complex n -dim'l vectors
 visualizations of vectors

ex: $\vec{v} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$



ex: $\vec{v} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$



Def: A matrix is said to be in reduced row echelon form (rref) if it satisfies:

(1) The leading nonzero entry in each row is 1.
 (called the leading one)

(2) If i th row has leading 1 at j th col, then $(i+1)^{th}$ row has leading 1 $> j$ th col

(3) Each col. containing a leading 1 has 0's in all other entries

Ex: $\begin{bmatrix} 0 & 0 & 1 & * & a & c & * & 0 \\ 0 & 0 & a & 0 & 1 & 0 & * & 0 \\ 0 & 0 & 0 & c & 0 & 1 & * & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ ex: $\begin{pmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 4 \end{pmatrix} = \text{rref}$

Ex: $\begin{bmatrix} 0 & 1 & 2 & 0 & 3 & 4 \\ 0 & 0 & 1 & 0 & 5 & 6 \\ 0 & 0 & 0 & 1 & 7 & 8 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} : \text{rref}$

Ex: $\begin{pmatrix} 0 & 1 & 2 & 0 \\ 1 & 0 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} : \text{not rref}$

Ex: $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 0 \end{bmatrix} : \text{not rref}$

we care about rref because we need to modify 3×4 matrix into another 3×4 in RREF.
we want to solve

$2x + 8y + 4z = 2$
 $2x + 5y + z = 5$
 $4x + 10y - z = 1$

Process modify def

- Three matrix operations (elementary row operations)
- 1.) Multiply/divide a row by a non-zero #.
 - 2.) Subtract a multiple of a row from another row. ± a row
 - 3.) Swap two rows.

$\left[\begin{array}{ccc|c} 2 & 8 & 4 & 2 \\ 2 & 5 & 1 & 5 \\ 4 & 10 & -1 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & * \\ 0 & 1 & 0 & * \\ 0 & 0 & 1 & * \end{array} \right]$

↓ solve this

$\left[\begin{array}{ccc|c} 2 & 8 & 4 & 2 \\ 2 & 5 & 1 & 5 \\ 4 & 10 & -1 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 4 & 2 & 1 \\ 2 & 5 & 1 & 5 \\ 4 & 10 & -1 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 4 & 2 & 1 \\ 0 & -3 & -3 & 3 \\ 4 & 10 & -1 & 1 \end{array} \right]$
 $2^{nd} \leftarrow 2^{nd} - 2 \cdot 1^{st}$

$1^{st} \leftarrow 1^{st} / 2$

$\left[\begin{array}{ccc|c} 1 & 4 & 2 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & -6 & -9 & -3 \end{array} \right] \xrightarrow{\begin{matrix} 1^{st} - 4 \cdot 2^{nd} \\ 3^{rd} + 6 \cdot 2^{nd} \end{matrix}} \left[\begin{array}{ccc|c} 1 & 0 & -2 & 5 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & -3 & -9 \end{array} \right] \xrightarrow{3^{rd} / -3} \left[\begin{array}{ccc|c} 1 & 0 & -2 & 5 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 3 \end{array} \right]$
 $1^{st} + 2 \cdot 3^{rd}$
 $2^{nd} - 3^{rd}$

$\rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 11 \\ 0 & 1 & 0 & -4 \\ 0 & 0 & 1 & 3 \end{array} \right]$
 This matrix means
 $x = 11$
 $y = -4$
 $z = 3$

Def: Given $a_{11}x + a_{12}y + a_{13}z = b_1$
 $a_{21}x + a_{22}y + a_{23}z = b_2$
 $a_{31}x + a_{32}y + a_{33}z = b_3$ $\rightarrow$ $\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$: augmented matrix

$\rightarrow$ $\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$: Coefficient matrix

$\star$ This method of reducing a given augmented matrix into a RREF matrix is called the $\star$ Gauss elimination process.

Given a system of lin eqs

$\Rightarrow$ augmented matrix

Gauss
 $\xrightarrow{\text{elim.}}$ matrix in RREF.

Thm. Given any matrix A

(1) Gauss elim. process works in finite steps.

$\star$ (2) The result RREF is unique.

The resulting matrix is called the only RREF of mat A , and denoted by $\text{rref}(A)$.